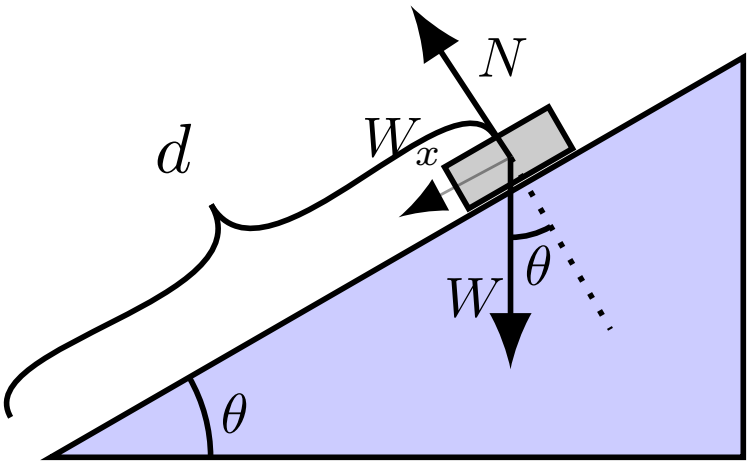
Motion Down A Ramp: Warm Up

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Consider a cart rolling down a slope with angle θ. The first thing we notice is that the forces are *not* balanced in this situation. Let's make our coordinate axis along the slope, such that the x axis is aligned with the slope going up, and the y axis is perpendicular to the slope. We have a Normal force, which is a third law pair canceling out the component $\perp$ to the slope of W. Thus there is no force in the y direction. This makes sense- there should be no net force in the y direction because the cart stays on the slope the entire time, and should have no y acceleration.

On the x axis, we just have the x component of W that didn't get canceled out, in the negative x direction. By Newton's second law, that means that there is some acceleration in the negative x direction.

$$F_{x,NET} = -W_x = ma_x \tag{1}$$
$$-mg\sin\theta = ma_x \tag{2}$$
$$-g\sin\theta = a_x \tag{3}$$

We also need to find the final velocity in terms of our known parameters (the angle θ and the distance traveled by the cart d, along with the mass of the cart and the acceleration from gravity.)

The energy conservation equation reads

$$E_{in} - E_{out} = E_{final} - E_{initial} \tag{4}$$

Meaning that the kinetic energy of the cart is increased/decreased by energy transferred in/out. We note that, if we ignore friction, that energy should only be transferred into the cart, since the only force acting on the cart is *in* the direction of movement.

We can use the energy transfer equation/ work-energy theorem to quantify this.

$$E_tran = F_{\parallel}d \tag{5}$$

Where $F_{\parallel}$ is the force component in the direction of motion. This is our energy in E_{in}.

We also need to calculate the initial and final kinetic energies. Note that the starting state is the cart released from rest, and so it has no kinetic energy.

At the end of the track we have some final velocity v. so

$$E_{final} = \frac{1}{2}mv^2 \tag{6}$$

And now, plugging in $E_{initial} = 0$, $E_{out} = 0$, $E_{in} = E_{tran} = F_{\parallel}d$ and $E_{final} = \frac{1}{2}mv^2$ to (**??**)

$$F_{\parallel}d = \frac{1}{2}mv^2 \tag{7}$$

The parallel force is just the force W_x

$$W_x d = \frac{1}{2}mv^2 \tag{8}$$

$$\frac{2W_x d}{m} = v^2 \tag{9}$$

$$2g\sin\theta d = v^2 \tag{10}$$

$$\sqrt{2g\sin\theta d} = v \tag{11}$$